Summary and Recommendation

Objective

The objective of this analysis is to explore customer churn patterns and identify customer characteristics associated with service discontinuation. The analysis focuses on demographic attributes, customer tenure, contract types, service subscriptions, and payment methods. Python libraries, including Pandas, NumPy, Matplotlib, and Seaborn, were used to clean the data, perform exploratory analysis, and create visualizations that help interpret customer behavior and identify potential areas for improving customer retention.

Key Insights & Findings

1. Overall Customer Churn

- The analysis reveals that **26.54% of customers have churned**, while 73.46% have remained with the company.
- The pie chart illustrates the percentage distribution of churned and retained customers, while the count plot compares the number of customers in both categories.
- **Implication:** The proportion of churned customers highlights the importance of understanding the factors associated with customer loss and identifying opportunities to improve retention.

2. Contract Type and Churn

- The contract-based count plot compares customer churn across month-to-month, one-year, and two-year contracts.
- Customers with month-to-month contracts are more likely to churn than customers with one-year or two-year contracts.
- **Implication:** Contract type appears to be an important factor associated with customer retention. The company could investigate incentives for longer-term contracts and improve retention efforts for month-to-month customers.

3. Churn by Tenure

- The tenure histogram compares customer tenure with churn status.
- The analysis indicates that customers who have used the company's services for a longer period tend to stay, whereas customers with very short tenure, particularly one or two months, appear more likely to churn.

- **Implication:** Early customer engagement may be important for retention. Better onboarding, timely support, and regular follow-ups could help identify potential customer concerns during the initial stages of service.

4. Senior Citizens and Churn

- The analysis uses a count plot and a stacked percentage bar chart to compare churn among senior and non-senior customers.
- The percentage-based visualization indicates that a comparatively greater proportion of senior citizens have churned.
- **Implication:** Senior-customer churn deserves further investigation to understand whether service experience, support requirements, pricing, or other factors may contribute to the observed difference.

5. Churn by Gender

- A count plot was created to compare churn among male and female customers.
- The visualization provides a comparison of churn patterns across both gender categories.
- **Implication:** Gender can be considered alongside other customer characteristics, but the chart alone does not establish it as a major factor influencing churn.

6. Churn by Internet and Other Services

- Nine count plots were created to examine PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, and StreamingMovies.
- The analysis observes that many retained customers have PhoneService, InternetService, particularly DSL, and OnlineSecurity enabled.
- The notebook also identifies noticeably higher churn among customers who do not use certain services, including OnlineBackup, TechSupport, and StreamingTV, or for whom these services are unavailable.
- **Implication:** Service subscriptions may be relevant to customer retention. Further analysis of churn percentages within each service category could help identify which services are most strongly associated with customer retention.

7. Payment Methods and Churn

- The payment-method count plot compares churned and retained customers using electronic checks, mailed checks, automatic bank transfers, and automatic credit-card payments.

- The analysis identifies electronic-check users as a group more likely to churn than customers using the other payment methods.
- **Implication:** The company could investigate the payment experience of electronic-check customers and evaluate whether billing improvements or alternative payment options could support customer retention.

Visualizations & Data Insights

The notebook includes several visualizations to explore customer churn from different perspectives:

- **Count Plot of Customer Churn:** Displays the number of churned and retained customers.
- **Pie Chart of Churn Distribution:** Shows that 26.54% of customers churned and 73.46% remained.
- **Count Plot by Gender:** Compares churn across male and female customers.
- **Stacked Percentage Bar Chart by Senior-Citizen Status:** Compares the proportions of churned and retained customers across senior and non-senior groups.
- **Tenure Histogram:** Illustrates the relationship between customer tenure and churn status.
- **Contract Type Count Plot:** Compares churn across month-to-month, one-year, and two-year contracts.
- **Service-Level Count Plots:** Examines churn across nine phone, internet, security, backup, device-protection, technical-support, and streaming variables.
- **Payment Method Count Plot:** Compares customer churn across four payment methods.

Together, these charts help identify differences in customer behavior and highlight the categories that may warrant further investigation. The overall churn chart and senior-citizen percentage chart provide percentage-based views, while most of the remaining charts display customer counts.

Recommendations

- **Improve early customer engagement:** Monitor new customers and provide timely support during the first few months of service.
- **Investigate contract-related churn:** Explore retention initiatives for month-to-month customers and evaluate the benefits of longer-term contract options.

- **Examine payment-method patterns:** Investigate the reasons behind higher observed churn among electronic-check users.
- **Evaluate service subscriptions:** Analyze churn percentages within each service category to understand the relationship between service adoption and retention.
- **Investigate senior-customer needs:** Examine the service experience and support requirements of senior customers to better understand their higher observed churn proportion.
- **Extend the analysis:** Calculate exact churn percentages for the different customer groups and investigate combinations of customer characteristics to obtain a more detailed understanding of churn patterns.

The project demonstrates the application of Python-based exploratory data analysis, data cleaning, descriptive statistics, and data visualization to extract meaningful insights from customer data. The findings provide a foundation for further investigation into customer retention and the development of more targeted churn-reduction strategies.